Design of 8x8 Bit Carry Lookahead Adder

Mohammed Obied
Computer Engineering Department
Birzeit University
Ramallah, Palestine
1210093@student.birzeit.edu

Malik Arqoup
Computer Engineering Department
Birzeit University
Ramallah, Palestine
1211686@student.birzeit.edu

Rami Arqoup
Computer Engineering Department
Birzeit University
Ramallah, Palestine
1210245@student.birzeit.edu

ABSTRACT

This paper provides the design and analysis of an 8x8 CMOS-based CLA. The design starts with a 2-bit CLA that is extended to a 4-bit CLA and further to an 8-bit CLA using hierarchical design techniques. The main challenge is to maximize performance improvements of the adder, specifically in terms of speed, power dissipation, and chip area. Reduced power consumption without sacrificing speed is achieved by using a supply voltage (VDD) of 0.8V. XOR, AND, OR, and inverters are incorporated, and sizing is optimized to balance NMOS and PMOS transistors. Ripple carry propagation is very fast in the CLA due to its structure, offering much better speed than traditional ripple-carry adders. Simulation results confirm the correctness of the design, making it suitable for low-power and high-speed digital systems.

INTRODUCTION

The need for high-speed digital circuits has increased the demand for efficient adder designs, which are essential components in many digital systems. Adders are used in arithmetic operations, signal processing, and data computation. This paper focuses on designing an 8x8 CLA adder with an emphasis on optimizing speed and power consumption within the CMOS technology platform.

The design begins with a 1-bit CLA, which serves as the foundation for larger designs. The 1-bit CLA is extended to a 2-bit CLA by cascading two 1-bit units. Next, a 4-bit CLA is developed using two 2-bit CLA modules and carry lookahead logic to achieve faster results. Finally, the 8-bit CLA is constructed by implementing two 4-bit CLAs while ensuring that carry inputs are smoothed to minimize arrival time.

To achieve maximum power efficiency without compromising speed, the design employs a VDD of 0.8V. The implementation includes digital logic gates such as XOR, AND, OR, and inverters. Each gate is carefully chosen and sized to meet the specifications of both PMOS and NMOS transistors in terms of area, speed, and power consumption.

The final design delivers a low-power, high-speed CLA that meets the application requirements of area-efficient processors, DSPs, and other high-performance systems.

IMPLEMENTATION AND DESIGN PROCEDURE

As mentioned, to implement this circuit firstly the basic gates were implemented.

I. Schematic and Layout

A. Basic Logic Gates

The XOR Gate: The XOR gate, which has been an important part of the CLA, was produced using CMOS logic. The whole design comprises fundamental gates, which are inverters and pass transistors. The sizes of both the PMOS and NMOS transistors were chosen to obtain the best performance. For the XOR gate the PMOS width was chosen to be 40 and the NMOS width was selected to be also 40 in order to have symmetrical rise and fall time. In the following section, the schematic of the XOR gate is provided as figure 1.

The AND Gate: The circuit for the AND gate in the CLA design was implemented by means of NAND gate with an inverted output. It simply has NMOS transistors connected in the series and PMOS transistors connected in parallel. For a 2-input AND gate, the width of the PMOS transistor is fixed to be 20 for each of the transistor and the NMOS width is double to 20 (20 x 1) because the transistor is in series. This leads to a configuration where the speed of the PMOS and NMOS are akin to that of the reference inverter speed. In the next figure, Figure 2 shows the schematic diagram of the AND gate circuit.

The OR Gate: The OR gate used in the CLA design was constructed using the NOR gate connected to an inverter. This core structure of the signal NOR gate itself is implemented using NMOS and PMOS transistors where the inputs of the gate are paralleled and the outputs are in serial. For a 2-input OR gate, the PMOS width remains at (20 X 2) 40 for each transistor and NMOS width is multiplied by two to serve as a series connection it becomes 10. This configuration makes the PMOS and NMOS capabilities equal to the reference inverter's speed. The OR gate is represented by the above said figure 3 and explained below that is;

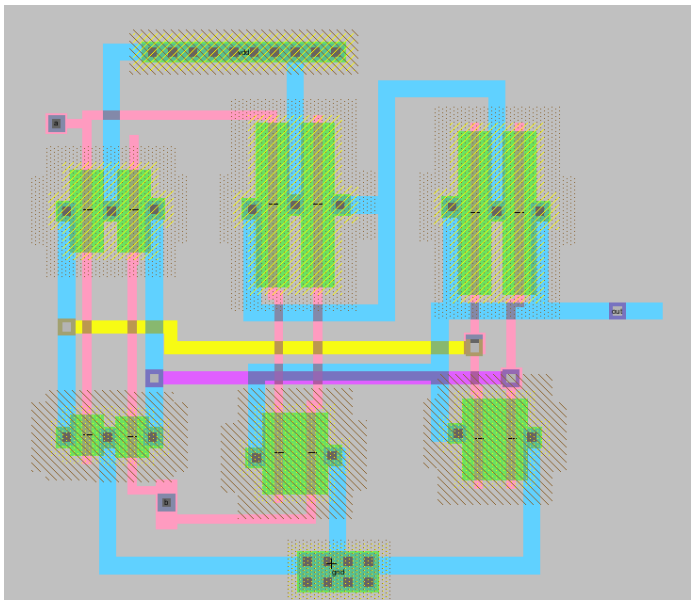


Fig. 2. xor GATE layout .

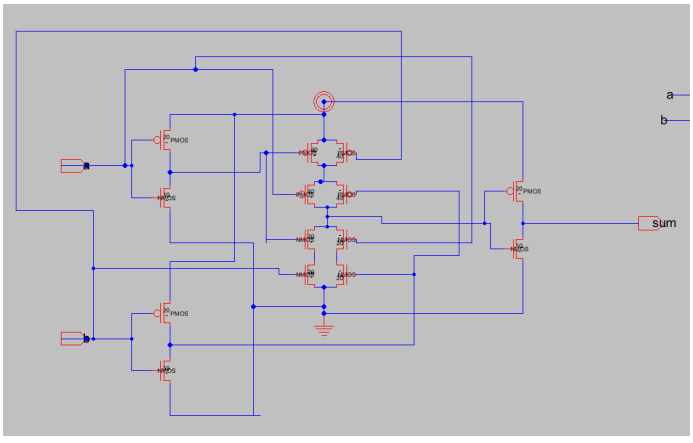


Fig. 1. XOR GATE Schematic .

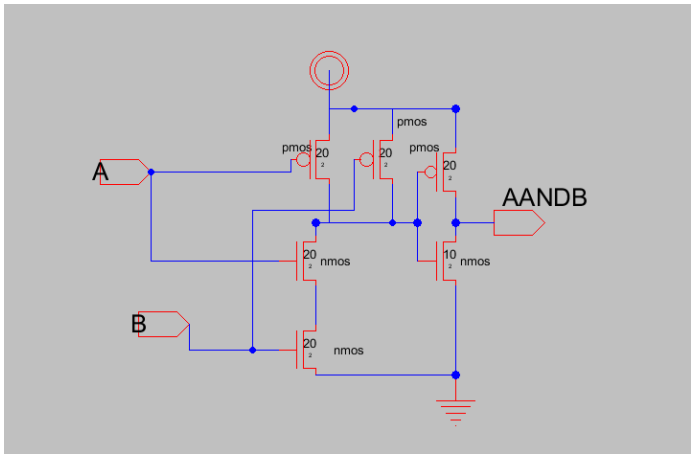


Fig. 3. AND GATE Schematic .

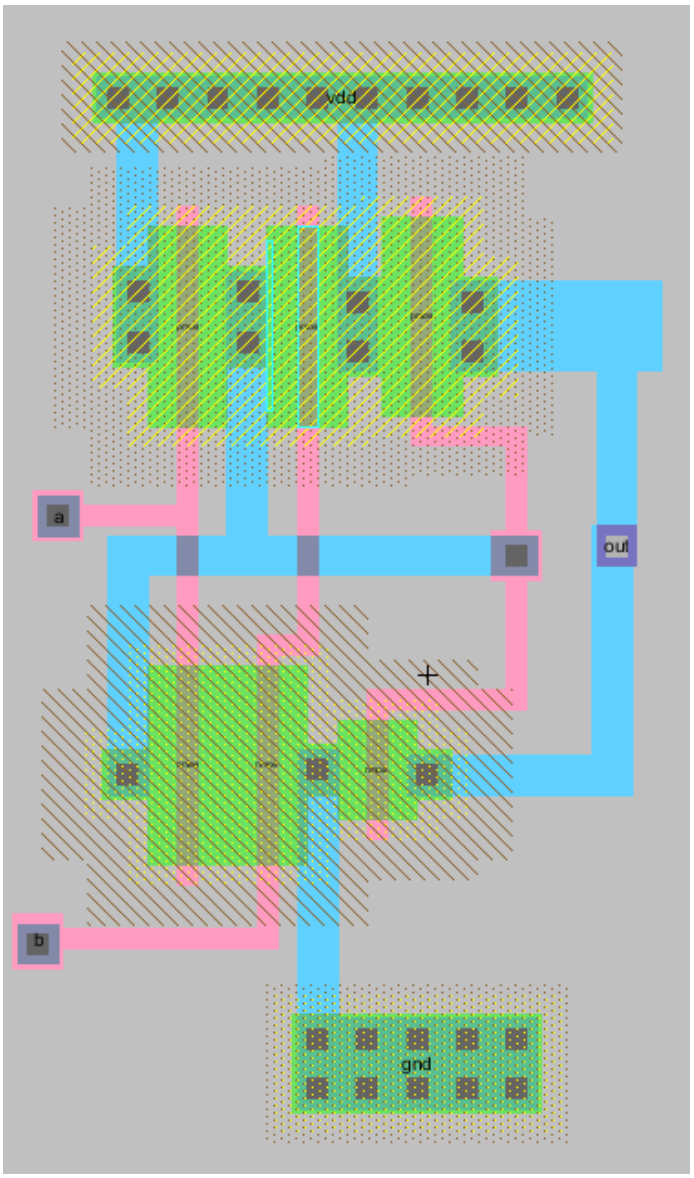


Fig. 4. and GATE layout .

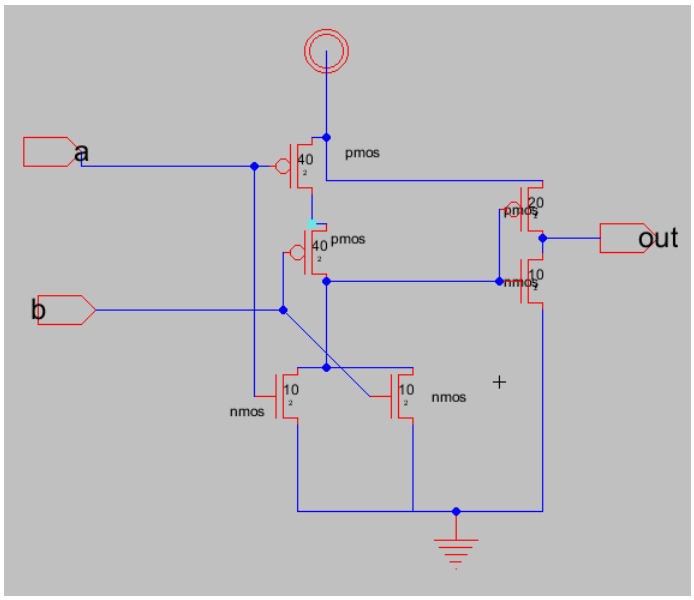


Fig. 5. OR GATE Schematic .

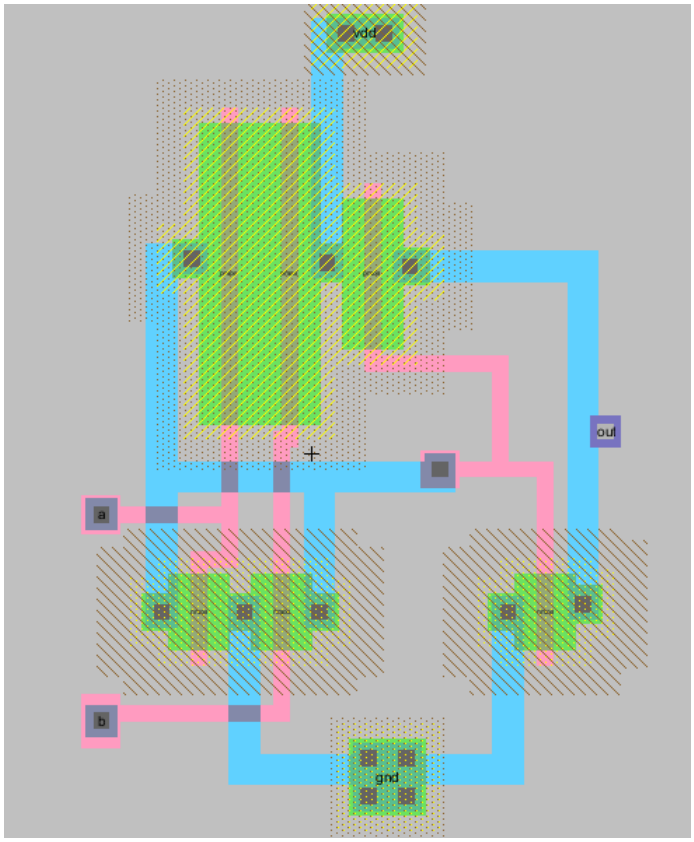


Fig. 6. OR GATE layout .

B. implement logic from the basics

II. ONE BIT ADDER

The circuit shown could be retrieved to represent a one bit adder, which is basic element in the construction of digital arithmetic circuits. It operates on three inputs: Carry-in inputs are a, b, and c, and the FA generates three output signals, sum, g (also known as the generate signal) and p (the propagate signal).

The sum output is obtain using the XOR operation hence it has the ability to produce the binary addition when considering the carry-in. The propagate signal p suggests whether or not a propagate can occur through out the bit, this is given as OR of the inputs a and b. Signal g is computed by performing an AND operation between 'a' and 'b,' where the generate signal, g, is used to decide whether a carry is generated at the present stage.

This circuit is aimed for single bit addition, but fulfills a very important role in digital systems for carry generation and propagation management.

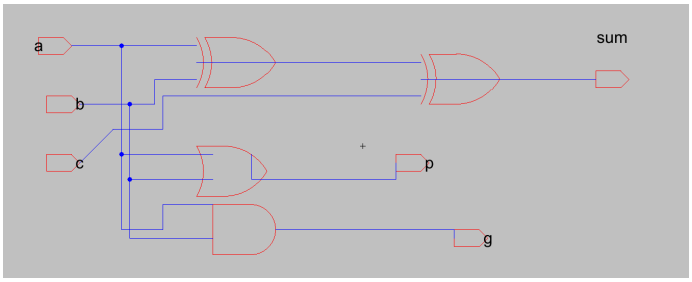


Fig. 7. one bit adder Schematic .

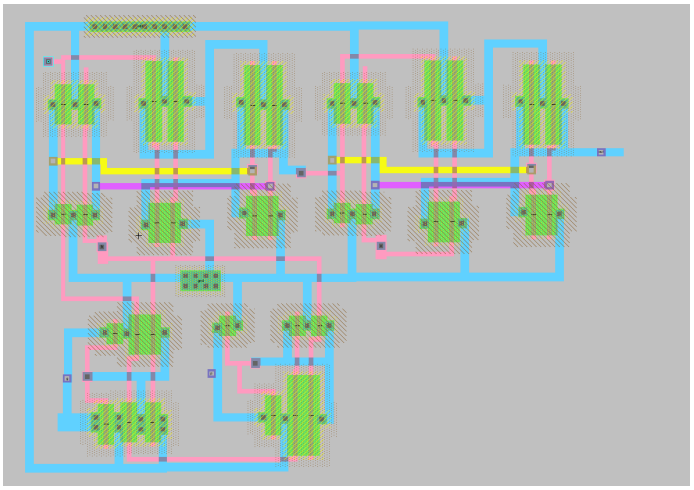


Fig. 8. one bit adder layout .

III. 2-BIT CARRY LOOKAHEAD ADDER

The 2-bit Carry Lookahead Adder abbreviated as CLA is ultra-fast digital circuit proved convenient for binary addition because it exhibits minimal propagation delay as opposed to ripple carry adders. The CLA attains this by precomputing the Generate (G) and Propagate (P) signals for each bit making it easier to make the carry signals at the same time. Represented in the schematic of CLA, the following is a logical structure made up of two full adder blocks to process two input bits; a 'a' and 'b' in addition to a carry in or 'Cin'. The outputs are a sum digit S0 and S1 digit, along with a carry out signal C1and c2. The used of AND, OR and XOR gates are effective to compute the sum and carry signals, based on the following equation

$$C_{\text{out}} = G + P \cdot C_{\text{in}} \tag{1}$$

. Employing CMOS technology, the layout design of a CLA showcases efficient PMOS and NMOS transistor placement with the interconnection traces running through the metal layers. This design achieves both speed and area improvements over a conventional design and the 2-bit CLA is an important building block for larger adders and a fundamental element in high-speed digital systems.

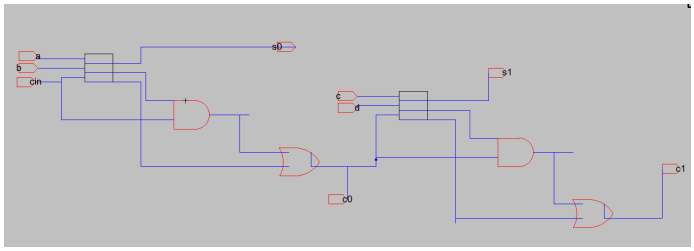


Fig. 9. 2-Bit Carry Lookahead Adder Schematic .

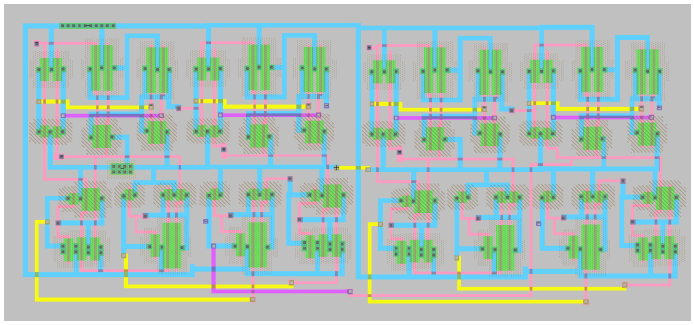


Fig. 10. 2-Bit Carry Lookahead Adder layout .

IV. 4-BIT CARRY LOOKAHEAD ADDER

Now, we go on to design a 4-bit adder with two 2-bit adders as the components needed. That calculation entails figuring out the sum outputs s0,s1,s2,s3 There are the carry outputs of the examiner and others as well as in The results of the examination have the carry outputs c1,c2,c3,c4

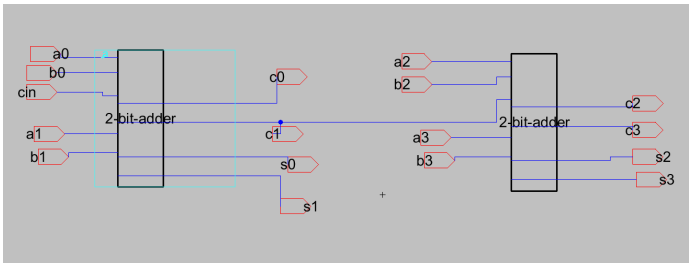


Fig. 11. 4-Bit Carry Lookahead Adder Schematic .

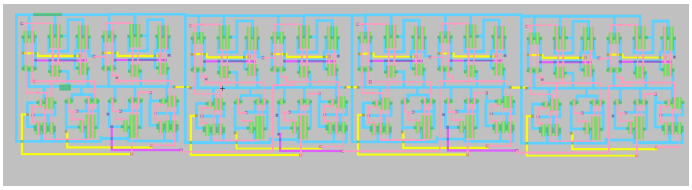


Fig. 12. 4-Bit Carry Lookahead Adder layout .

V. 8-Bit Carry Lookahead Adder

Now, we go on to design a 8-bit adder with two 4-bit adders as the components needed. That calculation entails figuring out the sum outputs s0,s1,s2,s3,s4,s5,s6,s7, There are the carry outputs of the examiner and others as well as in The results of the examination have the carry outputs c1,c2,c3,c4,c5,c6,c7,c8

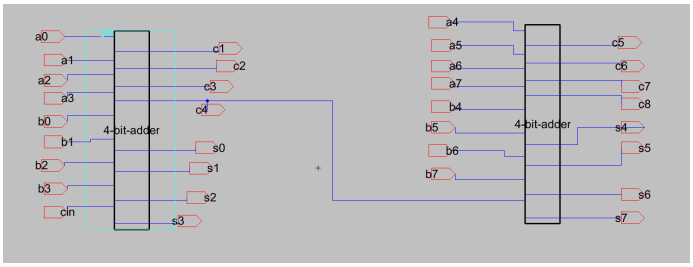


Fig. 13. 8-Bit Carry Lookahead Adder Schematic .

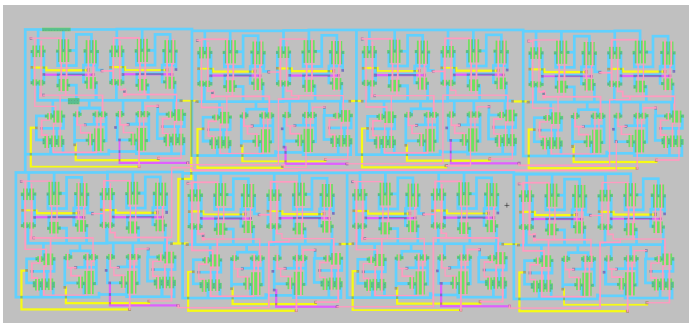


Fig. 14. 8-Bit Carry Lookahead Adder layout .

A. Simulation ,power and delay for 2-Bit Carry Lookahead Adder

By figure 16, we are able to analyze that the power consumption is nearly equal to (1e-7) and the delay is less than 1 nSec. This shows a marked increase thus implicating high performance, equally implying that the circuit is efficient enough a device to afford low power consumptions whilst affording high speed performance. Such metrics can give an insight why the current high-performance integrated electronic devices and energy-efficient systems design could be achieved.

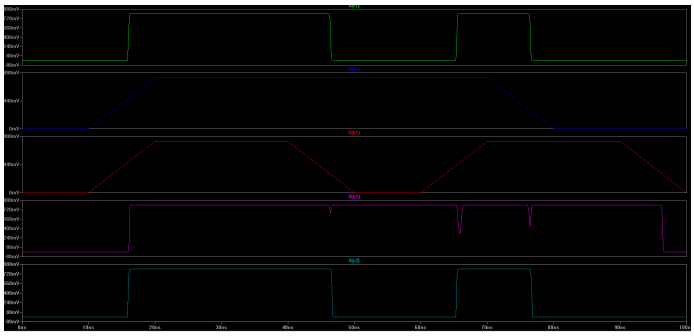


Fig. 15. 8-Bit Carry Lookahead Adder Schematic .

```
tf_s1=5.01659e-008 FROM 4.63267e-008 TO 9.64926e-008
tr_s1=1.73979e-010 FROM 1.60469e-008 TO 1.62209e-008
tf_c2=1.66938e-010 FROM 4.6554e-008 TO 4.672le-008
tr_c2=1.83092e-010 FROM 1.58707e-008 TO 1.60538e-008
prop_delays1=2.87209e-010 FROM 1.61126e-008 TO 1.63998e-008
prop_delayc2=4.06484e-010 FROM 1.594lle-008 TO 1.63476e-008
pavg=le-007 FROM 0 TO le-007
```

Fig. 16. 8-Bit Carry Lookahead Adder layout .

B. Simulation ,power and delay for 4-Bit Carry Lookahead Adder

By figure 18 ,analyzing the performance metrics of the 4-bit circuit, it is observed that the power consumption is nearly (1e-7), and the delay remains well below 1 nanoseconds. This demonstrates a noticeable enhancement in performance, emphasizing the circuit's capability to deliver high-speed operation while maintaining low power consumption. Such results confirm the circuit's efficiency and its suitability for use in energy-efficient designs and high-performance integrated electronic systems, making it an ideal candidate for modern digital applications.

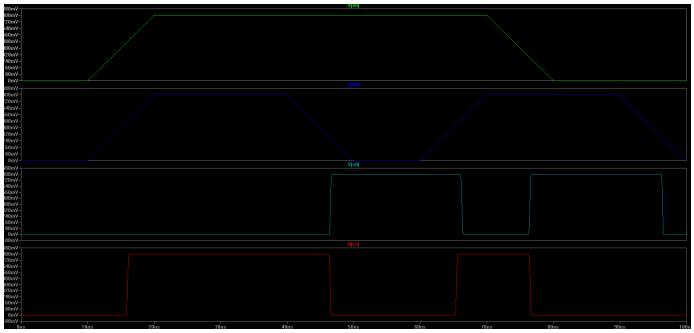


Fig. 17. 8-Bit Carry Lookahead Adder Schematic .

```
tf_s1=2.81798e-010 FROM 4.63092e-008 TO 4.6591e-008
tr_s1=1.73958e-010 FROM 1.6047e-008 TO 1.62209e-008
tf_c2=1.48692e-010 FROM 4.70398e-008 TO 4.71885e-008
tr_c2=1.83092e-010 FROM 1.58707e-008 TO 1.60538e-008
prop_delays1=2.87213e-010 FROM 1.61126e-008 TO 1.63998e-008
prop_delayc2=4.06479e-010 FROM 1.5941le-008 TO 1.63476e-008
pavg=le-007 FROM 0 TO le-007
```

Fig. 18. 8-Bit Carry Lookahead Adder layout .

C. Simulation ,power and delay for 8-Bit Carry Lookahead Adder

By figure 20 ,analyzing the performance metrics of the 8-bit circuit, it is observed that the power consumption is nearly (1e-7), and the delay remains well below 1 nanoseconds. This demonstrates a noticeable enhancement in performance, emphasizing the circuit's capability to deliver high-speed operation while maintaining low power consumption. Such results confirm the circuit's efficiency and its suitability for use in energy-efficient designs and high-performance integrated electronic systems, making it an ideal candidate for modern digital applications.

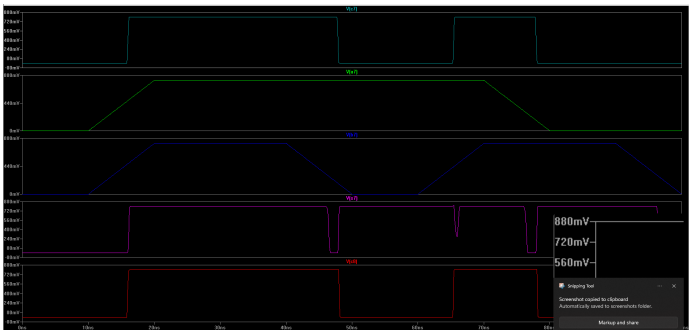


Fig. 19. 8-Bit Carry Lookahead Adder Schematic .

```
tf_s1=2.83615e-010 FROM 4.63092e-008 TO 4.65928e-008
tr_s1=1.66845e-010 FROM 1.6047e-008 TO 1.62138e-008
tf_c2=1.24859e-010 FROM 4.80017e-008 TO 4.81265e-008
tr_c2=1.83092e-010 FROM 1.58707e-008 TO 1.60538e-008
prop_delays7=1.2377e-009 FROM 4.52655e-008 TO 4.65032e-008
prop_delayc8=3.92111e-010 FROM 1.5941le-008 TO 1.63333e-008
pavg=le-007 FROM 0 TO le-007
```

Fig. 20. 8-Bit Carry Lookahead Adder layout .

CONCLUSION

In this work, we design and analyze the basic types of logical gates such as AND gate, OR gate, and XOR gate that forms the block for constructing complex but efficient digital circuits at their basic form. These gates were useful in in designing other complicated circuits such as the one-bit, two-bit, four-bit and eight-bit carry lookahead adder (CLA). It was found about the CLA design that many features outperformed the ripple-carry adders and also prohibited the propagation delay providing faster computing power by calculating the Generate (G) and propagate (P) signals before hand.

As seen in all designs, the analysis also maintained similar trends on all the performance measures.For all the various architectures–one-bit, two-bit, four-bit, and eight-bit–the power requirement stayed at around (1e-7) as the delay for each stage was less than 1 nanoseconds.served as the backbone for designing advanced arithmetic circuits such as the one-bit, two-bit, four-bit, and eight-bit carry lookahead adders (CLA). The CLA design exhibited significant improvements over traditional ripple-carry adders by addressing the propagation delay, enabling faster computations through the pre-computation of generate (G) and propagate (P) signals.

The analysis across all designs showed consistent results in terms of performance metrics. The power consumption for all configurations (one-bit, two-bit, four-bit, and eight-bit) remained approximately (1e-7), while the delay for each stage was consistently less than 1 nanosecond. These metrics show how integrated the CLA design is, in achieving high speed whilst having low energy requirements and is much faster and reliable than normal adders.

The improvement in speed and the decrease of power consumption were delivered because of operation with carry signals in parallel in the CLA while traditional designs have carry propagation bottleneck. Moreover, the scalability concern proved ineffective as the CLA was designed in such that the number of bits could range from one to eight with no impact on performance for demonstrating the FPGA capability for other extensive operations.

These results confirm that optimal digital circuit design is critical for fulfilling the needs of today's high-performance and power-constrained systems. Using low power consumption and high speed performance, the carry lookahead adder shows feasibility for implementation in processors, arithmetic logic units (ALUs), and other high-speed applications of fourth-generation computers. The success of this design underscores the importance of optimality in the complete design of the logic gates and the gross circuit designs toward further developments of the digital electronics.

REFERENCES

[1] "Electronics — Special Issue: Recent Advances in Digital Systems," Available: https://www.mdpi.com/2079-9292/13/18/3668, Accessed: Jan. 7, 2025.

[2] "Carry Look-Ahead Adder," GeeksforGeeks, Available: https://www.geeksforgeeks.org/carry-look-ahead-adder/, Accessed: Jan. 7, 2025.

[3] "Carry Lookahead Adders," ResearchGate, Available: https://www.researchgate.net/publication/262835866$_{C}arry_{L}ookahead_{A}dd$ Accessed: Jan. 7, 2025.